Lesson notes · for you

Draw it: charts

Data engineering · Level 1 · Lesson 3

Time
15 min

Materials
Squared paper, a pencil and a ruler; Coloured pencils; A phone or tablet with Darsi, for the lab

Learning goals

- Draw a pictogram and a bar chart.
- Read a bar chart: tallest, shortest, how many more.
- Check that a chart is fair.

Activity: Our fruit chart

1. Vote (3 min): The class votes for a favourite fruit, and two pupils keep a tally on the board.
2. Draw (6 min): Each pupil draws a bar chart on squared paper, with a title, labels and bars from zero.
3. Read (3 min): Find the tallest and the shortest bar, and work out how many more votes the winner got.
4. Check (3 min): Swap charts with a partner. Do the bars start at zero? Is every fruit shown?

Show one unfair chart on the board, with bars starting at 50, so pupils can spot the trick.

Discussion

- Why must bars start at zero?
- What does the key in a pictogram tell you?
- How did the red dots show us something was wrong?

Check yourself

- Draw a tiny bar chart with three bars, and write its title.

Try it this week
Turn your tally from the first lesson into a bar chart. Give it a title and labels, and start the bars at zero!

Worksheet

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Name _____ Class _____ Date _____

1 **Which of these charts is fair?**

- ◯ Only the biggest answers shown
- ◯ Bars that start at 50
- ◯ Bars that start at zero, with every answer shown

2 **In a bar chart, the tallest bar has the most votes.**

- ◯ True
- ◯ False

3 **Oranges got 12 votes and bananas got 7. Complete the sentence.**

Oranges got _____ more votes than bananas.

Word bank: 7 · 5 · 19

4 **Match each part of a chart to its job.**

The title	• • • •	Says which answer the bar is for
The label under a bar	• • • •	Says what the chart is about
The key in a pictogram	• • • •	Where every bar starts
Zero	• • • •	Says what one picture means

5 **Which of these make a chart unfair? Choose all that apply.**

- ☐ A title and labels
- ☐ Leaving out the small answers
- ☐ Bars that start at 50
- ☐ Bars of different widths

✎ **Have you seen a chart in a book, on TV or on a poster? Was it fair?**

Check your answers

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1 **Bars that start at zero, with every answer shown**

Bars that start at zero show the real differences, and every answer gets its bar, even the small ones.

2 **True**

The taller the bar, the more votes. That's why bars must start at zero: then their heights show the real numbers.

3 **Oranges got 5 more votes than bananas.**

12 take away 7 is 5. On a bar chart, that's the difference between the tops of the two bars.

4 **The title — Says what the chart is about**
The label under a bar — Says which answer the bar is for
The key in a pictogram — Says what one picture means
Zero — Where every bar starts

A title, labels, a key and bars from zero help everyone read a chart the same way.

5 **Bars that start at 50; Leaving out the small answers; Bars of different widths**

Bars from 50 make small differences look huge, missing answers hide the truth, and a wider bar looks bigger than it is.